

AI PM Interview Preparation Guide

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Complete project-by-project breakdown across:

Product Design · Product Development · Product Delivery · Cost & Pricing · Stakeholder Management · ML System Design · MLOps

Prepared for Senior / Group AI PM Interview Roles

Career Snapshot: 12+ years | \$15M+ revenue driven | FDA-approved AI product | 11th globally PAIP 2020 | ISB MBA Rank 1 | Co-founder \$0.5M in 6 months

Domains: iGaming · Healthcare & Digital Pathology · FinTech · Autonomous Systems · Retail · Commodity Markets

Core Stack: Python · PyTorch · TensorFlow · AWS SageMaker · Kafka · MLflow · Feast · LangGraph · LLMs · RAG · GNN · LSTM · U-Net · Mask R-CNN

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Onward Health · Cancer AI Research

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Project 1: Churn Prediction & Retention Engine

\$15M Annual Revenue Impact

One-line pitch: "I defined and delivered a real-time churn prediction platform that translated player behavioral signals into personalized retention interventions — driving \$15M in annual revenue uplift for a major Australian gaming client."

Product Design

	Detail
Problem Discovery	Interviewed ops, CRM, and marketing teams. Found 3 pain points: (1) no behavioral signal aggregation, (2) bonus spend was untargeted, (3) no LTV segmentation. Retention was reactive, not predictive.
Product Tension	Personalization vs. blanket bonuses. Resolved by building a tiered risk score (0–100) with SHAP-driven reasons — CRM team could act on WHY a player was flagged, not just that they were.
Core Principle	"Predict intent, not just behavior" — model player disengagement signals 7–14 days before churn event.
North Star Metric	LTV uplift per retained player cohort. Secondary: churn rate reduction %, bonus ROI improvement.
PRD Decisions	Personalized intervention > blanket bonus; tiered risk score > binary churn flag; SHAP explanations for CRM team to act on. V1 binary flag rejected by CRM → V2 tiered score adopted → V3 real-time Kafka triggers.

Product Development

Dimension	Detail
Feature Engineering	200+ behavioral features: session frequency, bet sizing variance, game diversity, deposit/withdrawal patterns, bonus response history, recency-frequency-monetary (RFM) signals.
Model Architecture	XGBoost for tabular signals + LSTM for sequential play patterns → ensemble with calibrated probability output → 0–100 risk score with SHAP attribution.
A/B Testing	Holdout group (no intervention) vs. treatment (personalized retention offer). Measured 30-day retention delta and LTV uplift per cohort. Stakeholder sign-off gate before each rollout phase.
Iteration	V1 → binary flag (rejected). V2 → tiered score + SHAP reasons (adopted). V3 → real-time scoring with Kafka event triggers for live session interventions.

Product Delivery

Dimension	Detail
MLOps Stack	Feast (feature store) → MLflow (experiment tracking + model registry) → Kafka (event streaming) → Airflow (nightly retraining pipeline with bias gate).
Latency Tiers	Real-time <50ms for live session triggers; near-real-time 5min Flink window for session-end scoring; nightly batch for CRM campaign targeting.
Monitoring	PSI drift monitoring on all input features; bias gate on retraining (model must pass fairness check before promotion); SHAP logged for 100% of high-risk flags.
Rollout	Shadow mode (2 weeks) → 10% traffic → full rollout with CRM integration. Stakeholder sign-off gate at each stage.

Cost & Pricing

Dimension	Detail
Build vs Buy	Evaluated Optimove, Symplify. Chose build: client data was proprietary and model needed embedding in real-time event pipeline. 3rd-party tools couldn't support <50ms latency requirement.
Infra Cost	Feast on AWS + SageMaker endpoints + Kafka MSK ≈ \$8K/month. Justified against \$15M revenue impact = 1800x ROI.
Optimization	Nightly batch for low-urgency segments (cheap); real-time only for high-value players (justified cost). Reduced inference cost by ~60% vs. all-real-time approach.

Stakeholder Management

Stakeholder	Approach
C-Suite	Monthly business reviews — presented LTV uplift, churn rate delta, revenue attribution. Framed AI as a revenue lever, not a tech project.
CRM Team	Co-designed intervention playbooks. They needed SHAP reasons to craft the right retention message per player segment.
Engineering	Weekly sprint reviews; defined API contracts for real-time scoring endpoint; managed latency SLA as a hard product requirement.
Compliance	UKGC responsible gambling rules — ensured model didn't target vulnerable players; added responsible gambling flag as a hard override on retention interventions.

ML System Design

Architecture: Event stream (Kafka) → Feature computation (Flink) → Feature store (Feast) → Model serving (SageMaker endpoint) → Score store (DynamoDB) → CRM API.

Training Pipeline: S3 data lake → Spark feature engineering → MLflow experiment tracking → SageMaker training job → model registry → A/B shadow deployment.

Serving: REST API with <50ms p99 latency; fallback to last-known score if model unavailable; circuit breaker pattern.

Retraining: PSI > 0.2 on key features OR weekly scheduled — whichever comes first. Bias gate: model must pass demographic parity check before promotion to production.

STAR Interview Answer:

S: Gaming client losing high-LTV players with no early warning system — retention was reactive.

T: Define and deliver a predictive retention platform with measurable revenue impact.

A: Replaced binary churn flag with 0–100 tiered risk score + SHAP reasons; built 3-tier latency architecture; co-designed CRM intervention playbooks; A/B tested personalized vs. blanket interventions.

R: \$15M annual revenue uplift; churn rate reduction across high-LTV player cohorts; CRM team adoption within 2 sprints of V2 launch.

Project 2: iGaming Fraud Detection Platform

7 Fraud Typologies · UKGC & MGA Compliant

One-line pitch: "I identified fraud as a top business risk through customer and ops interviews, prioritized 7 typologies by revenue impact, and led delivery of a production-grade multi-layered fraud detection platform with full regulatory compliance."

Product Design

Dimension	Detail
Problem Discovery	Customer + ops interviews revealed fraud costing 3–5% of GGR. No unified view — each team had siloed detection rules. Binary flags were causing false positives on high-value legitimate players.
Typology Prioritization	Mapped 7 fraud types by revenue impact × detection difficulty matrix. Bonus Abuse (high impact, medium difficulty) prioritized first. Currency Manipulation deprioritized to Phase 2.
Key PM Decision	Replaced binary fraud flag with 0–100 calibrated risk score + tiered actioning (REVIEW / RESTRICT / BLOCK). Reduced false positives on high-value players while catching fraud.
Core Principle	"Protect revenue, not just rules" — every model decision must be explainable to ops and auditable for regulators.

Product Development

Dimension	Detail
Ensemble Design	Isolation Forest (unsupervised anomaly) + Rule-Based flags + XGBoost (pseudo-label bootstrapped) + GNN (collusion ring detection) + LSTM (sequential bot behavior).
Why GNN	Collusion detection requires graph-level reasoning — player-to-player transaction networks. No tabular model can capture ring structures. GNN was the only viable approach.
Why LSTM	Bot behavior is sequential — inhuman click timing, session patterns. LSTM captures temporal dependencies that XGBoost misses.
Consensus Design	Weighted consensus amplifier — no single model can BLOCK a player. Requires weighted consensus across models to reduce false positives on legitimate high-value players. PM decision, not engineering default.

Product Delivery

Dimension	Detail
3-Tier Latency	

	Real-time (<50ms): deposit/withdrawal anomaly at transaction event. Near-real-time (5min Flink): session behavior scoring for bot detection. Nightly Airflow: GNN collusion ring analysis (computationally expensive, batch acceptable).
Regulatory	SAR filing automated within 24hrs of BLOCK decision. UKGC & MGA audit trail — every BLOCK decision logged with SHAP explanation, model version, and timestamp.
Drift Monitoring	PSI monitoring on all fraud features; auto-retrain trigger when distribution shifts (e.g., new fraud pattern emerges).

Cost & Pricing

Dimension	Detail
GNN Cost Mgmt	GNN inference is expensive — PM decision to run only in nightly batch for collusion detection. Acceptable latency tradeoff given collusion is a slow-burn fraud type.
Tiered Compute	Real-time tier uses lightweight models (rules + Isolation Forest). Heavy models (GNN, LSTM) run in batch. Reduced real-time infra cost by ~70%.
Regulatory ROI	UKGC non-compliance fines can reach £10M+. SAR automation + audit trail = compliance insurance. Framed to C-suite as risk mitigation with quantifiable downside protection.

Stakeholder Management

Stakeholder	Approach
Ops Team	Co-designed REVIEW/RESTRICT/BLOCK tiered actioning. Weekly calibration sessions to tune thresholds. They needed actionable tiers, not a raw score.
Legal/Compliance	Every BLOCK decision must have human-readable SHAP explanation for regulatory audit. Worked with legal to define SAR filing SLA (24hrs).
C-Suite	Presented fraud typology prioritization as a revenue protection roadmap — quantified expected savings per typology to justify build sequence.

ML System Design

Architecture: Kafka event stream → 3 parallel scoring pipelines (real-time rules, Flink session, nightly GNN) → weighted consensus aggregator → risk score store → ops dashboard + SAR system.

Graph Infrastructure: Neo4j/Neptune for player transaction graph — GNN trained on graph embeddings; updated nightly with new transaction edges.

Feature Store: Feast with 3 feature views: player-level (historical), session-level (real-time), network-level (graph features). Shared across fraud and churn models.

Explainability: SHAP values computed for every BLOCK decision; stored in audit log with model version, feature values, and decision timestamp.

STAR Interview Answer:

S: Fraud costing 3–5% GGR with no unified detection — siloed rules, binary flags causing false positives on VIP players.

T: Build a production-grade, multi-typology fraud platform with regulatory compliance.

A: Prioritized 7 typologies by revenue impact; designed weighted ensemble (Isolation Forest + XGBoost + GNN + LSTM); replaced binary flag with 0–100 tiered score; automated SAR filing; built 3-tier latency architecture.

R: Full UKGC & MGA compliance; SAR automation within 24hrs; reduced false positives on high-value players; tiered actioning adopted by ops team.

Project 3: AI Financial Wellness Chatbot — Old Mutual

FAIS-Compliant · 10x Cost Reduction · LangGraph RAG

One-line pitch: "I led end-to-end product and technical delivery of a FAIS-compliant conversational AI for Old Mutual — balancing 24/7 user experience against strict financial advice regulations, with a 10x LLM cost reduction through model splitting."

Product Design

Dimension	Detail
Problem	Old Mutual needed 24/7 financial guidance at scale — but FAIS regulations prohibit AI from making financial recommendations.
Core Tension	UX (helpful, personalized guidance) vs. Compliance (cannot recommend). Resolved with the "Show Options Rule" — AI presents 2–3 options with pros/cons, user makes the final choice.
Core Principle	"Show Options Rule" — AI presents choices, never recommends. Co-designed with legal team, not imposed by engineering.
Compliance as Feature	FAIS compliance positioned as a product differentiator in GTM: "the only AI financial assistant that's fully FAIS-compliant."

Product Development

Dimension	Detail
LangGraph Architecture	6-node stateful DAG: Intent Classification → Input Guardrail → RAG Retrieval → Response Generation → Suggestions → Output Guardrail. Stateful graph enables multi-turn conversation with context persistence.
RAG vs Fine-tuning	PM Decision: OM product docs update frequently. Fine-tuning = retraining on every doc update (expensive, slow). RAG = zero retraining cost, always up-to-date. Clear winner.
Model Split	Claude Sonnet 4.5 for complex intent + response; Haiku 3 for simple queries. ~10x cost reduction on general queries without quality degradation. Intent classifier routes ~70% to Haiku.
Grounding Threshold	0.7 cosine similarity — below threshold, AI says "I don't have enough information" rather than hallucinating. Compliance-critical design decision.

Product Delivery

Dimension	Detail
Infrastructure	Aurora RDS (pgvector for ANN search) + DynamoDB (session state) + AWS Bedrock (LLM serving) + Gravitee API Gateway.

Security Stack	Ping Federate OAuth2 (authentication) + Gravitee (rate limiting, API security) + dual-layer Bedrock Guardrails (PII masking, denied topics, prompt injection prevention).
DynamoDB Decision	Redis is faster but more expensive and complex at pilot scale. DynamoDB = cost-effective, serverless, sufficient for pilot. Revisit at scale — documented as a known trade-off.
Rollout	Internal pilot (OM employees) → limited customer beta → full rollout. Each stage: defined acceptance criteria (response accuracy, guardrail trigger rate, escalation rate, NPS).

Cost & Pricing

Dimension	Detail
LLM Cost	Sonnet 4.5 (\$3/M tokens) for complex queries; Haiku 3 (\$0.25/M tokens) for simple queries. ~70% of queries routed to Haiku → ~10x blended cost reduction.
RAG Infra Cost	pgvector on Aurora RDS (no separate vector DB license) + Cohere embeddings API (pay-per-use). Total RAG infra ~\$500/month at pilot scale.
Pricing Model	SaaS per-conversation pricing for client. Pilot at fixed fee, scale at per-MAU pricing. Justified against cost of human financial advisors (\$50–100/consultation).

Stakeholder Management

Stakeholder	Approach
Legal/FAIS	Weekly compliance reviews throughout development. "Show Options Rule" co-designed with legal. Legal sign-off was a hard gate before any user-facing release.
OM Product Owner	Monthly product reviews; defined acceptance criteria together (response accuracy, guardrail trigger rate, user satisfaction NPS).
Engineering	LangGraph state management was complex — weekly architecture reviews to manage node dependencies and error handling in the stateful graph.
End Users	User testing sessions with OM customers — discovered "explain this in simple terms" was a core need. Added to V2 roadmap.

ML System Design

LangGraph Architecture: 6-node directed acyclic graph with stateful context. Each node is a discrete LLM call or retrieval step. State persists across turns via DynamoDB session store.

RAG Pipeline: Document ingestion → Cohere embedding → pgvector index on Aurora RDS → ANN search (cosine similarity, threshold 0.7) → top-k chunks → LLM context injection.

Dual Guardrail Design: Input guardrail (before RAG) — blocks off-topic, harmful, PII-containing queries. Output guardrail (after generation) — checks for compliance violations, PII leakage, denied topics.

Fallback Design: If grounding score < 0.7 → "I don't have enough information, please speak to an advisor." Human escalation path always available.

STAR Interview Answer:

S: Old Mutual needed 24/7 AI financial guidance but FAIS regulations prohibit AI recommendations.

T: Deliver a compliant, scalable conversational AI that provides genuine user value without violating regulations.

A: Defined "Show Options Rule" with legal; built LangGraph 6-node stateful agent; chose RAG over fine-tuning; implemented model split for 10x cost reduction; dual-layer guardrails for compliance.

R: FAIS-compliant product delivered; 10x LLM cost reduction; successful pilot with OM employees; legal sign-off achieved.

Project 4: Game Recommendation + Bonus Analysis + Demand Forecasting

4-Product AI Portfolio · GTM + KPI Ownership

One-line pitch: "I drove GTM and iterative product development for 4 AI products in the gaming portfolio — each with defined KPIs, A/B testing frameworks, and stakeholder sign-off gates."

Product Design & Development

Product	Algorithm	PM Decision
Game Recommendation Engine	Collaborative filtering + content-based hybrid; real-time personalization based on play history and player segment	Hybrid over pure CF to handle cold-start for new games; KPI: click-through rate + session depth increase
Bonus Analysis System	Statistical analysis for bonus ROI attribution; causal inference to separate bonus effect from organic retention	Defined bonus ROI as the north star — not just redemption rate. Prevented over-spending on low-LTV players.
Demand Forecasting	ARIMA/Prophet/LSTM ensemble for multi-horizon forecasting; external regressors (events, seasonality)	3 forecast horizons (1-day, 7-day, 30-day) for different operational decisions; MAPE <10% as acceptance criterion

Product Delivery & MLOps

Architecture: Modular microservices per product; shared Feast feature store across all 4 products; shared A/B testing framework.

Delivery: Each product had independent KPIs, GTM strategy, and stakeholder sign-off gates. Iterative roadmap with defined success metrics per release.

Portfolio Management: Managed 4 products simultaneously — prioritized by revenue impact and engineering capacity. Weekly portfolio review with stakeholders.

Project 5: FDA-Approved AI Medical Scanner

FDA Approval · Full Product Lifecycle Ownership

One-line pitch: "I owned the full product lifecycle — requirements, clinical validation, regulatory documentation — for an AI-intelligent digital scanner, achieving FDA approval while managing cross-functional alignment between engineering, clinical, and regulatory stakeholders."

Product Design

Dimension	Detail
Problem	Clinical imaging workflows required a smarter scanner that could integrate AI-based image analysis at the point of capture — reducing downstream processing time and diagnostic errors.
Regulatory as Design Constraint	FDA 510(k) pathway requirements shaped every product decision — from data collection protocols to model validation methodology to documentation standards.
North Star	FDA approval as the ultimate product acceptance criterion. Every feature, validation, and documentation decision was evaluated against this gate.

Product Development & Delivery

Dimension	Detail
Algorithm	CNN-based image analysis pipeline for clinical image interpretation; validation against clinical ground truth with defined sensitivity/specificity benchmarks.
Full Lifecycle	Data pipeline → model development → clinical validation → regulatory documentation → FDA submission → approval. Each phase had defined exit criteria.
Hardware Integration	AI model integrated with scanner hardware — real-time inference at point of capture. Latency and power constraints shaped model architecture choices.

Stakeholder Management

Stakeholder	Approach
Clinical Team	Co-defined validation protocols and ground truth labeling standards. Clinical sign-off at each validation milestone.
Regulatory Team	Weekly documentation reviews; ensured all model decisions were traceable and auditable for FDA submission.
Engineering	Managed hardware-software integration; latency SLAs as hard product requirements given real-time clinical use.

STAR Interview Answer:

S: Clinical imaging needed AI integration at point of capture — no existing product combined scanner hardware with real-time AI analysis.

T: Own full product lifecycle from concept to FDA approval.

A: Defined requirements with clinical partners; built CNN-based analysis pipeline; managed clinical validation; led regulatory documentation; coordinated engineering, clinical, and regulatory teams.

R: FDA approval achieved — a rare outcome for an AI medical product. Product positioned as a market differentiator for Photon360.

Project 6: Autonomous Parking — Volley Automation (Series-A, SF)

300+ Hours Robot Downtime Saved · Series-A Startup

One-line pitch: "I defined the data product strategy for a Series-A robotics startup — designing corporate data flow integrating sales, engineering, and marketing to save 300+ hours of robot downtime and improve operational SLAs."

Product Design & Development

Dimension	Detail
Problem	Robotics startup had siloed data across sales, engineering, and marketing — no unified view of robot performance, maintenance needs, or operational bottlenecks.
Algorithm	Camera-sensor fusion for spatial awareness; object detection/tracking for parking garage navigation; predictive maintenance models on robot operational data.
Data Strategy	Designed corporate data flow integrating 3 data streams (sales pipeline, engineering telemetry, marketing metrics) into a unified operational dashboard.
Investor Alignment	Data strategy aligned to Series-A investor narrative — demonstrated operational efficiency and scalability metrics that investors needed to see.

Project 7: EdTech Fraud Detection

\$1M Annual Savings

One-line pitch: "I scoped the fraud problem through user and ops research, prioritized fraud typologies by financial impact, and delivered an AI fraud detection system generating \$1M in annual savings for an online education platform."

Product Design & Development

Dimension	Detail
Discovery	User and ops research to scope fraud problem — identified 3 primary fraud typologies: fake account creation, payment fraud, and credential sharing.
Algorithm	Anomaly detection + behavioral pattern analysis; rule-based + ML hybrid for fraud typology coverage. Prioritized by financial impact × detection feasibility.
Delivery	End-to-end pipeline: data ingestion → feature engineering → model → alerting system → ops dashboard. Defined \$1M annual savings as the success metric.

Projects 8–10: Digital Pathology AI — Mayo Clinic

68% Time Reduction · 99.9% Processing Speedup · 66% Cost Reduction

One-line pitch: "I partnered with Mayo Clinic to define AI solution scope, success metrics, and delivery milestones — delivering a WSI digitization pipeline, tumor segmentation system, and TIL assessment tool that transformed clinical pathology workflows."

Project 8: WSI Digitization Pipeline

Dimension	Detail
Problem	Biopsy tissue digitization was a bottleneck — multi-hour manual process, error-prone, expensive. Pathologists were the constraint in the diagnostic pipeline.
Algorithm	AI-driven WSI pipeline; image preprocessing + quality assessment; dimensional reduction for gigapixel-scale images. DICOM ingestion → OpenSlide reader → adaptive tiling → GPU batch inference.
PM Metrics	Primary: 68% reduction in digitization time. Secondary: patient metadata mapping error rate (target: 0%). Tertiary: pathologist adoption rate.
Business Impact	Transformed multi-hour manual process into near-automated clinical workflow. Directly improved lab throughput and pathologist productivity.

Project 9: Tumor Segmentation (U-Net + Mask R-CNN)

Dimension	Detail
Algorithm	U-Net (semantic segmentation, pixel-level) + Mask R-CNN (instance segmentation, individual tumor regions). Ensemble for higher recall — clinical cost of false negative >> false positive.
Processing Speed	Hours → <1 second per WSI (99.9% reduction). Enabled real-time pathologist workflow integration.
TAM Expansion	Strategic AI integration reduced scanner cost by 66% vs. competitors — PM decision to position as a mid-tier hospital product. Expanded TAM 3–4x.
ML System Design	Gigapixel image → tiling (256×256 patches, 50% overlap) → parallel GPU inference → prediction stitching → whole-slide output. TensorRT optimization for inference speed.

Project 10: TIL Assessment — Breast Cancer Immunotherapy

Dimension	Detail
Clinical Impact	TIL grading is a critical biomarker for immunotherapy treatment targeting. Automated grading directly informs oncologist treatment decisions.
Algorithm	

Cell detection → TIL classification → density mapping at WSI level. Biomarker quantification pipeline with clinical-grade accuracy benchmarks.

PM KPI

Reduced time-to-insight for oncologists — clinical impact as the product KPI. Measured as: time from biopsy to TIL score available to oncologist.

Stakeholder Management — Mayo Clinic Projects

Stakeholder	Approach
Mayo Clinic	Defined success metrics jointly. Monthly clinical review meetings — pathologists reviewed model outputs and provided ground truth corrections for retraining.
Pathologists	Co-designed UI overlay — pathologists needed AI segmentation overlaid on WSI viewer, not a separate report. Reduced adoption friction significantly.
Hospital Procurement	Presented 66% cost reduction + clinical validation data. ROI framed as labor savings + diagnostic accuracy improvement.

STAR Interview Answer (Mayo Clinic Portfolio):

S: Biopsy digitization was a multi-hour bottleneck; tumor segmentation took hours per slide; TIL grading was manual and inconsistent.

T: Define AI solution scope, success metrics, and delivery milestones in partnership with Mayo Clinic.

A: Built WSI pipeline (68% time reduction); U-Net + Mask R-CNN tumor segmentation (99.9% speedup); automated TIL grading; patient metadata automation; co-designed pathologist UI overlay.

R: 68% digitization time reduction; <1 second per WSI tumor segmentation; 66% scanner cost reduction enabling TAM expansion to mid-tier hospitals.

Projects 11–14: Cancer AI Research Portfolio

11th Globally PAIP 2020 · \$2M Seed Funding · 8+ CV Products

One-line pitch: "I translated unmet diagnostic needs into an AI product roadmap — prioritizing 8 computer vision projects by clinical impact, data availability, and fundraising narrative — directly contributing to \$2M in seed funding and a globally ranked research outcome."

Project 11: PAIP 2020 — MSI Prediction (Ranked 11th Globally)

Dimension	Detail
Strategy	PAIP 2020 wasn't just a competition — it was a product strategy. Ranked 11th globally = credibility signal to investors, clinical partners, and talent.
Algorithm	U-Net (tissue segmentation) + EfficientNetB1 (MSI/MSS classification) ensemble. Image stitching strategy for 1 lakh × 1 lakh pixel images → 70% accuracy improvement. Managed ~200GB colorectal cancer tissue dataset.
ML System Design	WSI → adaptive tiling → patch extraction → parallel GPU inference → prediction stitching → ensemble output. Stitching: 50% overlap tiles, majority voting on overlapping predictions.
PM Outcome	Published research paper. Positioned Onward Health as credible AI research partner. Directly contributed to \$2M seed funding narrative.

Projects 12–14: Computer Vision Portfolio

Project	Algorithm	PM Framing
Skin Lesion Diagnosis	MobileNetV2 + DenseNet201 + InceptionV3 ensemble; cosine similarity image retrieval; >93% accuracy across 6 classes	Pathologist decision-support tool (not replacement); accuracy benchmark co-defined with clinical partners; retrieval network for similar case lookup
PD-L1 Detection	Patch extraction → cell detection → PD-L1 positive identification in lung adenocarcinoma; multi-stage pipeline	Top 3 at IIM-B Conference; presented to clinical and investor audiences; part of fundraising narrative
MSI/MSS Classification	VGG19 + VGG16 ensemble; microsatellite instability/stability in colorectal carcinoma; 82% accuracy	Directly informs treatment selection decisions; clinical partner-defined accuracy benchmarks
Breast Cancer Metastases + Mitosis + Tumor Budding	Detection and classification pipelines for lymph node metastases, mitosis counting, tumor budding identification	Each scoped with clinical partners; defined accuracy benchmarks before development; part of diagnostic platform portfolio

Roadmap Prioritization Framework

Prioritization Matrix: Each of 8 projects scored on: (1) Clinical Impact, (2) Data Availability, (3) Fundraising Narrative Strength, (4) Technical Feasibility.

PD-L1 and MSI prioritized because they directly inform treatment decisions — highest clinical + investor appeal.

Portfolio ROI: Cost of building 8 projects $\approx$ \$200K in team + compute. Leverage = \$2M seed funding. 10x return on research investment.

STAR Interview Answer:

S: Startup needed to demonstrate AI diagnostic capability to investors and clinical partners to secure seed funding.

T: Build a portfolio of 8 CV projects that demonstrates diagnostic accuracy, clinical relevance, and scalable platform potential.

A: Prioritized projects by clinical impact $\times$ fundraising narrative; led PAIP 2020 (11th globally); built skin lesion, PD-L1, MSI/MSS, and metastasis detection systems; presented at IIM-B Conference.

R: \$2M seed funding secured; 11th globally at PAIP 2020; published research paper; Top 3 at IIM-B Conference.

Projects 15–16: Healthcare & Infrastructure AI

30% Patient Inflow Increase · \$1M/yr GMR Airport

Project 15: OPD Patient Flow Forecasting

Dimension	Detail
Problem	India's largest eye care chain had unpredictable patient inflow — leading to over/understaffing, long wait times, and poor patient experience.
Algorithm	SARIMA + XGBoost ensemble for multi-horizon forecasting (1-day, 7-day, 30-day). Feature engineering: appointment history, seasonal patterns, local events, doctor availability, holiday calendar.
Product Design	Designed for hospital ops managers, not data scientists. Simple interface: "Expected patients tomorrow: 340 (± 20).". Scheduling recommendation engine + ops dashboard.
PM Outcome	30% patient inflow increase measured over 3-month post-deployment period. Directly managed C-suite client relationships. Doubled company's project portfolio.
Monitoring	Forecast accuracy tracked daily (MAPE). Alert if MAPE > 15% — triggers manual review and model recalibration.

Project 16: CCTV Analytics — GMR Airport

Dimension	Detail
Problem	GMR Airport security team manually reviewed CCTV footage — labor-intensive, reactive, not scalable. No automated vehicle intelligence.
Algorithm	YOLO-based real-time object detection → number plate OCR (OpenCV + Tesseract) → vehicle counting via centroid tracking → alert system for anomalies.
ML System Design	RTSP stream → frame sampling (5fps) → YOLO inference (GPU) → DeepSORT tracking → plate OCR → event store → dashboard WebSocket push. On-premise GPU server with TensorRT optimization.
Pricing Model	Per-camera licensing model — scalable with client infrastructure. On-premise deployment + annual maintenance fee. GMR = anchor client for airport vertical.
PM Outcome	\$1M annual monetization for GMR Airport. Reduced security team redundant workload. Became company-level showcase project for infrastructure vertical.

Project 17: Stock Momentum Prediction Engine

Wolf of Dalal Street Award · Rising Star Award

Dimension	Detail
Algorithm	Time series analysis (ARIMA, momentum indicators); statistical hypothesis testing for market inefficiency detection; backtesting framework for strategy validation.
PM Framing	Delivered precise prediction ranges (not point estimates) for portfolio performance — uncertainty quantification as a product feature. Bridged traditional finance with data science.
Outcome	Enhanced portfolio performance for institutional and retail clients. Recognized with "Wolf of Dalal Street" award. Foundation for quantitative thinking that evolved into full data science expertise.

Project 18: Cross-Exchange Arbitrage (LME + CME)

Systematic Arbitrage · LME + CME · Early Quantitative Foundation

Dimension	Detail
Algorithm	Rule-based + statistical models for arbitrage window detection; position sizing optimization; execution timing models. Cross-exchange price inefficiency detection in metals and energy markets.
PM Framing	Self-directed product owner — defined strategy, risk parameters, and execution rules. Early-stage quantitative thinking that seeded the data science career.
Foundation	Built proprietary rule engine; real-time price feed processing. Developed systematic thinking about signal detection, risk management, and execution optimization — directly applicable to ML system design.

Appendix: Interview Frameworks & Quick Reference

How to Frame Every Project in an Interview

- 1. Problem Discovery:** "The problem I identified through discovery was..." — show you did customer/ops research, not just built what was asked.

2. Product Tension: "The key product tension was X vs Y..." — UX vs. compliance, false positives vs. catch rate, cost vs. latency.

3. Core Principle: "My core product principle was..." — Show Options Rule, Predict Intent Not Behavior, Protect Revenue Not Just Rules.

4. North Star Metric: "The north star metric I defined was..." — LTV uplift, digitization time reduction, churn rate delta, \$1M savings.

5. Trade-off Made: "The key trade-off I made was..." — RAG vs. fine-tuning, build vs. buy, real-time vs. batch, GNN vs. tabular.

6. Outcome Measured: "The outcome I measured was..." — always quantified, always tied to business impact.

ML System Design — Universal Framework

Layer	Your Strongest Examples
Data	200GB WSI pipeline (PAIP), Kafka event streaming (Fraud/Churn), DICOM ingestion (Mayo Clinic)
Features	Feast feature store (shared across Churn + Fraud), 200+ behavioral features, graph embeddings (GNN), patch-level histopathology features
Model	Ensemble design rationale (why GNN for collusion, why LSTM for bots, why RAG over fine-tuning, why U-Net)